

The Sigmoid function is:

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

Sigmoid derivative via chain rule:

$$\begin{aligned}\frac{d}{dx} \sigma(x) &= \frac{d}{dx} \left[\frac{1}{1+e^{-x}} \right] \\&= \frac{d}{dx} (1+e^{-x})^{-1} \\&= -1 * (1+e^{-x})^{-2} (-e^{-x}) \\&= \frac{-e^{-x}}{(1+e^{-x})^2} \\&= \frac{1}{1+e^{-x}} \cdot \frac{e^{-x}}{1+e^{-x}} \\&= \frac{1}{1+e^{-x}} \cdot \frac{e^{-x} + (1-1)}{1+e^{-x}} \\&= \frac{1}{1+e^{-x}} \cdot \frac{(1+e^{-x}) - 1}{1+e^{-x}} \\&= \frac{1}{1+e^{-x}} \left[\frac{1+e^{-x}}{1+e^{-x}} - \frac{1}{1+e^{-x}} \right] \\&= \frac{1}{1+e^{-x}} \left[1 - \frac{1}{1+e^{-x}} \right] \\&= \sigma(x) (1 - \sigma(x))\end{aligned}$$